



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 18/07/2019*

PROBLEM

We observe the process $X[n] = S[n] + W[n]$, where the signal of interest $S[n]$ is modelled as a Moving Average (MA) process of order 1, i.e. $S[n] = I[n] - \alpha \cdot I[n-1]$, where $|\alpha| < 1$ and $I[n]$ is the zero-mean innovation process with variance σ_I^2 ; $W[n]$ is additive white noise, independent of $S[n]$, with power σ_w^2 .

(1) Derive the system function $L(z)$ of the innovation filter of the MA(1) process $S[n]$, its zeros and poles, and determine for which values of α it is a low-pass, high-pass, or band-pass filter. Derive the autocorrelation function (ACF) $R_S[m] = E\{S[n]S[n+m]\}$ and the power spectral density (PSD) $S_S(e^{j2\pi f}) = FT\{R_S[m]\}$ of the process $S[n]$ and plot them for $\sigma_I^2 = 1$ and $\alpha = 0.9$. Finally, derive the signal to noise ratio $\gamma \triangleq E\{S^2[n]\} / E\{W^2[n]\}$.

(2) Derive the optimal linear predictor at step L of order $P=2$ of the process $S[n]$, i.e. the linear minimum mean square error (LMMSE) estimator of $S[n+L-1]$ given the noisy observed data $X[n-1]$ and $X[n-2]$; derive explicitly the solution in the case $L=1$ and in the case $L=2$, and express it as a function of α and γ . Calculate numerically the impulse response of the linear predictor assuming $\sigma_w^2 = 1$, $\sigma_I^2 = 1$, and $\alpha = 0.9$.

(3) Derive its mean square error (MSE) $\sigma_\varepsilon^2 \triangleq E\{\varepsilon^2[n+L-1]\} = E\left\{\left(S[n+L-1] - \hat{S}[n+L-1]\right)^2\right\}$ in the case $L=1$ and in the case $L=2$, assuming $\sigma_w^2 = 1$, $\sigma_I^2 = 1$ and $\alpha = 0.9$, and comment the results.



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

Solution.

$$(1) S[n] = I[n] - \alpha I[n-1]$$

$$S(z) = I(z) - \alpha z^{-1} I(z)$$

$$L(z) = \frac{S(z)}{I(z)} = 1 - \alpha z^{-1} = \frac{z - \alpha}{z}$$

The innovation filter has 1 zero at $z = \alpha$ and 1 pole at $z = 0$. Hence, the innovation filter is a high-pass filter if $\alpha > 0$ and a low-pass filter $\alpha < 0$. The process $S[n]$ is consequently a high-pass process if $\alpha > 0$ and a low-pass process $\alpha < 0$. If $\alpha = 0$ we have $S[n] = I[n]$, so it is a white process. It cannot be a pass-band process for any value of α .

The ACF and the PSD are given by

$$\begin{aligned} R_s[m] &= E\{S[n]S[n+m]\} = E\{(I[n] - \alpha \cdot I[n-1])(I[n+m] - \alpha \cdot I[n+m-1])\} \\ &= R_I[m] - \alpha \cdot R_I[m-1] - \alpha \cdot R_I[m+1] + \alpha^2 \cdot R_I[m] \\ &= (1 + \alpha^2) \cdot \sigma_I^2 \delta[m] - \alpha \cdot \sigma_I^2 \delta[m-1] - \alpha \cdot \sigma_I^2 \delta[m+1] \end{aligned}$$

$$R_s[m] = \begin{cases} (1 + \alpha^2) \cdot \sigma_I^2, & m = 0 \\ -\alpha \cdot \sigma_I^2, & m = \pm 1 \\ 0, & |m| \geq 2 \end{cases}$$

$$\begin{aligned} S_s(e^{j2\pi f}) &= FT\{R_s[m]\} = \sum_{m=-\infty}^{+\infty} R_s[m] e^{-j2\pi f m} = \sigma_I^2 \sum_{m=-\infty}^{+\infty} [(1 + \alpha^2) \cdot \delta[m] - \alpha \cdot \delta[m-1] - \alpha \cdot \delta[m+1]] e^{-j2\pi f m} \\ &= \sigma_I^2 (1 + \alpha^2 - \alpha e^{-j2\pi f} - \alpha e^{j2\pi f}) = \sigma_I^2 (1 + \alpha^2 - 2\alpha \cos(2\pi f)) \end{aligned}$$

For $\sigma_I^2 = 1$ and $\alpha = 0.9$ the process is an high-pass process.

$$\gamma \triangleq E\{S^2[n]\} / E\{W^2[n]\} = \frac{(1 + \alpha^2) \cdot \sigma_I^2}{\sigma_W^2}$$

$$(2) \hat{S}[n+L-1] = h[1]X[n-1] + h[2]X[n-2] + b$$

Since all the processes are zero mean we have that: $E\{\varepsilon[n+L-1]\} = 0 \Rightarrow b = 0$.



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$E\{\varepsilon[n+L-1]X[n-m]\} = 0, \quad m=1,2$$

$$E\left\{\left(S[n+L-1]-\hat{S}[n+L-1]\right)X[n-m]\right\} = 0, \quad m=1,2$$

$$E\left\{\left(S[n+L-1]-h[1]X[n-1]-h[2]X[n-2]\right)X[n-m]\right\} = 0, \quad m=1,2$$

$$R_{sx}[-m-L+1]-h[1]R_x[-m+1]-h[2]R_x[-m+2] = 0, \quad m=1,2$$

$$R_s[m+L-1]-h[1]R_x[m-1]-h[2]R_x[m-2] = 0, \quad m=1,2$$

since:

$$\begin{aligned} R_{sx}[-m-L+1] &= E\{S[n+L-1]X[n-m]\} = E\{S[n+L-1](S[n-m]+W[n-m])\} \\ &= E\{S[n+L-1]S[n-m]\} = R_s[m+L-1] \end{aligned}$$

Consequently, the discrete-time Wiener-Hopf equation is given by:

$$\begin{cases} h[1]R_x[0] + h[2]R_x[-1] = R_s[L] \\ h[1]R_x[1] + h[2]R_x[0] = R_s[L+1] \end{cases}$$

$$\begin{bmatrix} R_x[0] & R_x[-1] \\ R_x[1] & R_x[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[L] \\ R_s[L+1] \end{bmatrix}$$

$$R_x[m] = R_s[m] + R_w[m] = R_s[m] + \sigma_w^2 \delta[m]$$

$$\begin{bmatrix} (1+\alpha^2) \cdot \sigma_l^2 + \sigma_w^2 & -\alpha\sigma_l^2 \\ -\alpha\sigma_l^2 & (1+\alpha^2) \cdot \sigma_l^2 + \sigma_w^2 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[L] \\ R_s[L+1] \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} (1+\alpha^2) \cdot \sigma_l^2 + \sigma_w^2 & -\alpha\sigma_l^2 \\ -\alpha\sigma_l^2 & (1+\alpha^2) \cdot \sigma_l^2 + \sigma_w^2 \end{bmatrix}^{-1} \begin{bmatrix} R_s[L] \\ R_s[L+1] \end{bmatrix}$$

$$\gamma = \frac{(1+\alpha^2) \cdot \sigma_l^2}{\sigma_w^2} \Rightarrow \frac{\sigma_l^2}{\sigma_w^2} = \frac{\gamma}{1+\alpha^2}$$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$L=1$:

$$\begin{aligned}
 \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} &= \frac{1}{\sigma_w^2} \begin{bmatrix} (1+\alpha^2) \cdot \frac{\sigma_l^2}{\sigma_w^2} + 1 & -\alpha \frac{\sigma_l^2}{\sigma_w^2} \\ -\alpha \frac{\sigma_l^2}{\sigma_w^2} & (1+\alpha^2) \cdot \frac{\sigma_l^2}{\sigma_w^2} + 1 \end{bmatrix}^{-1} \cdot \sigma_w^2 \begin{bmatrix} -\alpha \frac{\sigma_l^2}{\sigma_w^2} \\ 0 \end{bmatrix} = \begin{bmatrix} \gamma+1 & -\frac{\alpha}{1+\alpha^2} \gamma \\ -\frac{\alpha}{1+\alpha^2} \gamma & \gamma+1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} -\frac{\alpha}{1+\alpha^2} \gamma \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} (1+\alpha^2)(1+\gamma) & -\alpha\gamma \\ -\alpha\gamma & (1+\alpha^2)(1+\gamma) \end{bmatrix}^{-1} \cdot \begin{bmatrix} -\alpha\gamma \\ 0 \end{bmatrix} \\
 &= \frac{1}{(1+\alpha^2)^2(1+\gamma)^2 - (\alpha\gamma)^2} \begin{bmatrix} (1+\alpha^2)(1+\gamma) & \alpha\gamma \\ \alpha\gamma & (1+\alpha^2)(1+\gamma) \end{bmatrix} \begin{bmatrix} -\alpha\gamma \\ 0 \end{bmatrix} \\
 &= \frac{1}{(1+\alpha^2)^2(1+\gamma)^2 - (\alpha\gamma)^2} \begin{bmatrix} -(1+\alpha^2)(1+\gamma)\alpha\gamma \\ -(\alpha\gamma)^2 \end{bmatrix}
 \end{aligned}$$

$$h[1] = -\frac{(1+\alpha^2)(1+\gamma)\alpha\gamma}{(1+\alpha^2)^2(1+\gamma)^2 - (\alpha\gamma)^2}, \quad h[2] = -\frac{(\alpha\gamma)^2}{(1+\alpha^2)^2(1+\gamma)^2 - (\alpha\gamma)^2}$$

Assuming $\sigma_w^2 = 1$, $\sigma_l^2 = 1$, and $\alpha = 0.9$: $h[1] = -0.3569$, $h[2] = -0.1143$

$L=2$:

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} \gamma+1 & -\frac{\alpha}{1+\alpha^2} \gamma \\ -\frac{\alpha}{1+\alpha^2} \gamma & \gamma+1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(3) $\hat{S}[n+L-1] = h[1]X[n-1] + h[2]X[n-2]$



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

$$\begin{aligned}\sigma_\varepsilon^2 &= E\{\varepsilon^2[n+L-1]\} = E\left\{\left(S[n+L-1] - \hat{S}[n+L-1]\right)S[n+L-1]\right\} \\ &= E\left\{\left(S[n+L-1] - h[1]X[n-1] - h[2]X[n-2]\right)S[n+L-1]\right\} \\ &= R_s[0] - h[1]R_s[L] - h[2]R_s[L+1]\end{aligned}$$

$$R_s[0] = (1 + \alpha^2) \cdot \sigma_t^2 = 1.81, \quad R_s[1] = -\alpha \cdot \sigma_t^2 = -0.9, \quad R_s[2] = 0$$

$$L=1: \quad \sigma_\varepsilon^2 = R_s[0] - h[1]R_s[1] - h[2]R_s[2] = 1.81 - (-0.3569) \cdot (-0.9) = 1.4888$$

$$L=2: \quad \sigma_\varepsilon^2 = R_s[0] - h[1]R_s[2] - h[2]R_s[3] = 1.81$$